PRE-PH.D. COURSE WORK – SYLLABUS

(Aligned with UGC Ph.D. Regulations, 2022)

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1. GENERAL FRAMEWORK (As per UGC Regulations, 2022)

The Pre-Ph.D. Course Work shall be mandatory for all admitted Ph.D. scholars as per the **UGC (Minimum Standards and Procedure for Award of Ph.D. Degree) Regulations, 2022**.

Minimum duration: **One Semester**

- Minimum credits: **12 Credits**
- Medium of instruction and examination: English
- Minimum passing requirement: 55%

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2. STRUCTURE OF PRE-PH.D. COURSE WORK

Paper	Title	Credits	Max. Marks	Duration
I	Research Methodology	04	100	3 Hours
II	Computer Applications / Research Tools	02	100	3 Hours
III	Domain-Specific Course	04	100	3 Hours
IV	Research and Publication Ethics (RPE)	02	100	3 Hours
Total		**12 Credits**		

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3. DETAILED SYLLABUS

PAPER – I: RESEARCH METHODOLOGY

Credits: 04 | **Maximum Marks:** 100 | **Examination Duration:** 3 Hours

Course Objectives:
To equip research scholars with a comprehensive understanding of research concepts, research design, data collection methods, sampling techniques, hypothesis testing, and data analysis tools relevant to interdisciplinary research.

UNIT – I: INTRODUCTION (14 Hours)

Meaning, objectives and motivation of research; types of research; research approaches; significance of research; research methods and research methodology; scientific methods; research process; criteria of good research; interdisciplinary research and its implications; research problem—meaning, selection, necessity and techniques of problem definition.

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UNIT – II: DATA COLLECTION (14 Hours)

Primary data collection methods—observation, interview; questionnaires and schedules; secondary data collection; selection of appropriate data collection methods (quantitative and qualitative); case study method; experimental methods; questionnaire design; interviewing techniques; survey versus experiment; laboratory experimental studies; content analysis; tools of data collection and their applications.

UNIT – III: SAMPLE DESIGN (10 Hours)

Sampling concepts; characteristics and types of sample design; probability and non-probability sampling; merits and demerits of sampling; random sampling techniques; experimental procedures; control observations; sampling errors—Type I and Type II.

UNIT – IV: HYPOTHESIS (10 Hours)

Meaning and formulation of hypothesis; hypothesis testing procedures; power of test; parametric tests; testing of means; testing of correlation coefficients; limitations of hypothesis testing.

UNIT – V: PROCESSING AND ANALYSIS OF DATA (12 Hours)

Measures of central tendency and dispersion; correlation and regression analysis; one-sample and two-sample tests; chi-square, t-test, F-test and Z-test; standard deviation—applications and limitations; ANOVA and ANCOVA; discriminant, cluster and factor analysis; qualitative data analysis; presentation of data using tables and graphs; estimation of means and proportions; parametric and non-parametric decision making.

Suggested Readings:

1. Kothari, C.R. (2019). *Research Methodology: Methods and Techniques*. New Age International.
2. Creswell, J.W. (2018). *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches*. Sage.
3. Bryman, A. (2016). *Social Research Methods*. Oxford University Press.
4. Saunders, M., Lewis, P., & Thornhill, A. (2019). *Research Methods for Business Students*. Pearson.
5. Yin, R.K. (2018). *Case Study Research and Applications*. Sage.

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PAPER – II: COMPUTER APPLICATIONS / RESEARCH TOOLS

Credits: 02 | **Maximum Marks:** 100 | **Examination Duration:** 3 Hours

Course Objectives:
To develop essential computer skills and familiarity with digital tools required for research documentation, data analysis, and academic communication.

UNIT – I: COMPUTER FUNDAMENTALS (06 Hours)

Computer basics; data representation; input and output devices; memory; generations and classification; computer languages; operating systems—types, services and components; computer networks; Internet, WWW and E-mail.

UNIT – II: MS-OFFICE AND ITS APPLICATIONS (05 Hours)

Windows file handling; MS Word—formatting and macros; MS Excel—functions and formulas; MS PowerPoint—presentation design and delivery.

UNIT – III: WORD PROCESSING PACKAGE (07 Hours)

Research paper preparation; document design; cover letter and resume; title page, tables, charts and watermarks; mail merge; professional newsletter creation.

UNIT – IV: SPREADSHEET PACKAGE (07 Hours)

Spreadsheet creation and formatting; data analysis; charts; pivot tables; worksheet consolidation and printing.

UNIT – V: APPLICATION OF INTERNET (05 Hours)

Internet resources for research; DOAJ; e-journals; e-libraries; INFLIBNET; EBSCO Host; online academic databases.

Suggested Readings:
1. Rajaraman, V. (2014). *Fundamentals of Computers*. PHI Learning.
2. Shelly, G.B., Cashman, T.J., & Vermaat, M.E. *Microsoft Office Series*. Cengage.
3. Field, A. (2018). *Discovering Statistics Using SPSS*. Sage.
4. Anderson, D.R., Sweeney, D.J., & Williams, T.A. (2017). *Statistics for Business and Economics*. Cengage.
5. Jain, R. (2015). *Information Technology for Management*. McGraw-Hill.

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PAPER – III: DOMAIN-SPECIFIC COURSE

Credits: 04 | **Maximum Marks:** 100 | **Examination Duration:** 3 Hours

Course Objectives:
To provide advanced subject-specific knowledge and research orientation relevant to the scholar's proposed area of research.

Syllabus:
Available in concern Department/School as per UGC Regulations, 2022.

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PAPER – IV: RESEARCH AND PUBLICATION ETHICS (RPE)

Credits: 02 | **Maximum Marks:** 100 | **Examination Duration:** 3 Hours

UNIT – I: PHILOSOPHY AND ETHICS *(03 Hours)*

Definition, nature and scope of philosophy; branches of philosophy; ethics and moral philosophy; moral judgments and reactions.

UNIT – II: SCIENTIFIC CONDUCT *(05 Hours)*

Ethics in science and research; intellectual honesty; research integrity; scientific misconduct—fabrication, falsification and plagiarism (FFP); redundant publications; selective reporting and misrepresentation of data.

UNIT – III: PUBLICATION ETHICS *(07 Hours)*

Publication ethics—importance and best practices; standards and guidelines (COPE, WAME); conflicts of interest; authorship and contributorship; publication misconduct; identification, complaints and appeals; predatory journals and publishers.

UNIT – IV: OPEN ACCESS PUBLISHING *(04 Hours)*

Open access initiatives; SHERPA/RoMEO; tools for identifying predatory journals; journal finder tools—JANE, Elsevier Journal Finder, Springer Journal Suggestion.

UNIT – V: PUBLICATION MISCONDUCT *(04 Hours)*

Group discussion on ethical issues; FFP; authorship disputes; conflicts of interest; plagiarism detection tools such as Turnitin, Drillbit and open-source software.

UNIT – VI: DATABASES AND RESEARCH METRICS *(07 Hours)*

Citation databases—Web of Science and Scopus; journal metrics—Impact Factor, SNIP, SJR, CiteScore; author metrics—h-index, g-index, i10-index, altmetrics.

Suggested Readings:

1. University Grants Commission (2019). *Research and Publication Ethics (RPE) – Course Material*.
2. Chaddah, P. (2018). *Ethics in Competitive Research*. Springer.
3. National Academies of Sciences (2009). *On Being a Scientist*. National Academies Press.
4. Resnik, D.B. (2011). *What is Ethics in Research & Why is it Important*. NIEHS.
5. Beall, J. (2012). *Predatory Publishers are Corrupting Open Access*. Nature.

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